Polynomial Freiman-Ruzsa covering in bounded-torsion groups

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Editorial note. Source-organized author material only. All mathematical source spans are retained unchanged. Blueprint lean/proves/uses metadata is hidden as formatting only; numbering is local to this material wrapper.

Editorial note. Ordinary cited prerequisites and concise author arguments are accepted as such. No independent mathematical certification or public ID allocation is claimed.

Editorial note. Formatting repair after actual visual inspection: A4 landscape with18mm margins prevents long original displayed formulas being clipped.

Allowed standard prerequisites (editorial).

- Standard finite Shannon entropy identities, conditional entropy and finite abelian group algebra are cited in their original Blueprint formulations.
- Named prior PFR/entropy or BSG results are allowed prerequisites; their original statements are retained where used, without demanding recursively expanded proofs.

1 Proof of PFR

Lemma 1 (Ruzsa covering lemma)**.** *If A, B are finite non-empty subsets of a group G, then A can be covered by at most $|A+B|/|B|$ translates of $B-B$.*

Proof. Cover A greedily by disjoint translates of B. □

Lemma 2. *If $A \subset \mathbf{F}_2^n$ is non-empty and $|A+A| \leq K|A|$, then A can be covered by at most $K^{13/2}|A|^{1/2}/|H|^{1/2}$ translates of a subspace H of $\mathbf{F}_2^n$ with*

$$|H|/|A| \in [K^{-11}, K^{11}]. \tag{1}$$

Proof. Let U_A be the uniform distribution on A (which exists by Theorem 7), thus $\mathbb{H}[U_A] = \log|A|$ by Theorem 8. By Theorem 6 and the fact that $U_A + U_A$ is supported on $A+A$, $\mathbb{H}[U_A+U_A] \leq \log|A+A|$. By Theorem 35, the doubling condition $|A+A| \leq K|A|$ therefore gives

$$d[U_A; U_A] \leq \log K.$$

By Theorem 52, we may thus find a subspace H of $\mathbb{F}_2^n$ such that

$$d[U_A; U_H] \leq \tfrac{1}{2} C' \log K \tag{2}$$

with $C' = 11$. By Theorem 38 we conclude that

$$|\log|H| - \log|A|| \leq C' \log K,$$

proving (1). From Theorem 35, (2) is equivalent to

$$\mathbb{H}[U_A - U_H] \leq \log(|A|^{1/2}|H|^{1/2}) + \tfrac{1}{2}C' \log K.$$

By Theorem 9 we conclude the existence of a point $x_0 \in \mathbb{F}_p^n$ such that

$$p_{U_A - U_H}(x_0) \geq |A|^{-1/2}|H|^{-1/2}K^{-C'/2},$$

or equivalently

$$|A \cap (H + x_0)| \geq K^{-C'/2}|A|^{1/2}|H|^{1/2}.$$

Applying Theorem 1, we may thus cover A by at most

$$\frac{|A + (A \cap (H + x_0))|}{|A \cap (H + x_0)|} \leq \frac{K|A|}{K^{-C'/2}|A|^{1/2}|H|^{1/2}} = K^{C'/2+1}\frac{|A|^{1/2}}{|H|^{1/2}}$$

translates of

$$\big(A \cap (H + x_0)\big) - \big(A \cap (H + x_0)\big) \subseteq H.$$

This proves the claim. □

Theorem 3 (PFR). *If $A \subset \mathbf{F}_2^n$ is non-empty and $|A + A| \leq K|A|$, then A can be covered by most $2K^{12}$ translates of a subspace H of $\mathbf{F}_2^n$ with $|H| \leq |A|$.*

Proof. Let H be given by Theorem 2. If $|H| \leq |A|$ then we are already done thanks to (1). If $|H| > |A|$ then we can cover H by at most $2|H|/|A|$ translates of a subspace H' of H with $|H'| \leq |A|$. We can thus cover A by at most

$$2K^{13/2}\frac{|H|^{1/2}}{|A|^{1/2}}$$

translates of H', and the claim again follows from (1). □

2 Shannon entropy inequalities

Random variables in this paper are measurable maps $X : \Omega \to S$ from a probability space Ω to a measurable space S, and called S-valued random variables. In many cases we will assume that singletons in S are measurable. Often we will restrict further to the case when S is finite with the discrete σ-algebra, which of course implies that S has measurable singletons.

Definition 4 (Entropy). *If X is an S-valued random variable, the entropy $\mathbb{H}[X]$ of X is defined*

$$\mathbb{H}[X] := \sum_{s \in S} \mathbb{P}[X = x] \log \frac{1}{\mathbb{P}[X = x]}$$

with the convention that $0 \log \frac{1}{0} = 0$.

Lemma 5 (Entropy and relabeling). *(i) If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, and $Y = f(X)$ for some injection $f : S \to T$, then $\mathbb{H}[X] = \mathbb{H}[Y]$.*

(ii) If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, and $Y = f(X)$ and $X = g(Y)$ for some functions $f : S \to T$, $g : T \to S$, then $\mathbb{H}[X] = \mathbb{H}[Y]$.

Proof. Expand out both entropies and rearrange. □

Lemma 6 (Jensen bound). *If X is an S-valued random variable, then $\mathbb{H}[X] \leq \log |S|$.*

Proof. This is a direct consequence of Theorem 49 and Jensen's inequality. □

Lemma 7 (Uniform distributions exist). *Given a finite non-empty subset H of a set S, there exists a random variable X (on some probability space) that is uniformly distributed on H.*

Proof. Direct construction. □

Lemma 8 (Entropy of uniform random variable, II). *If X is uniformly distributed on H, then, then $\mathbb{H}[X] = \log |H|$.*

Proof. Direct computation. □

Lemma 9 (Bounded entropy implies concentration). *If X is an S-valued random variable, then there exists $s \in S$ such that $\mathbb{P}[X = s] \geq \exp(-\mathbb{H}[X])$.*

Proof. We have

$$\mathbb{H}[X] = \sum_{s \in S} \mathbb{P}[X = s] \log \frac{1}{\mathbb{P}[X = s]} \geq \min_{s \in S} \log \frac{1}{\mathbb{P}[X = s]}$$

and the claim follows. □

We use X, Y to denote the pair $\omega \mapsto (X(\omega), Y(\omega))$.

Lemma 10 (Commutativity and associativity of joint entropy). *If $X : \Omega \to S$, $Y : \Omega \to T$, and $Z : \Omega \to U$ are random variables, then $\mathbb{H}[X, Y] = \mathbb{H}[Y, X]$ and $\mathbb{H}[X, (Y, Z)] = \mathbb{H}[(X, Y), Z]$.*

Proof. Set up an injection from (X, Y) to (Y, X) and use Theorem 5 for the first claim. Similarly for the second claim. □

Note: it may happen that E has zero measure. In which case, the ambient probability measure should be replaced with a zero measure. (In our formalization we achieve this by working with arbitrary measures, and normalizing them to be probability measures if possible, else using the zero measure. Conditioning is also formalized using existing Mathlib definitions.)

Definition 11 (Conditional entropy). *If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, the conditional entropy $\mathbb{H}[X|Y]$ is defined as*

$$\mathbb{H}[X|Y] := \sum_{y \in Y} \mathbb{P}[Y = y] \mathbb{H}[(X|Y = y)].$$

Lemma 12 (Conditional entropy and relabeling). *If $X : \Omega \to S$, $Y : \Omega \to T$, and $Z : \Omega \to U$ are random variables, and $Y = f(X, Z)$ almost surely for some map $f : S \times U \to T$ that is injective for each fixed U, then $\mathbb{H}[X|Z] = \mathbb{H}[Y|Z]$.*

Similarly, if $g : T \to U$ is injective, then $\mathbb{H}[X|g(Y)] = \mathbb{H}[X|Y]$.

Proof. For the first part, use Theorem 11 and then Theorem 5. The second part is a direct computation. □

Lemma 13 (Chain rule). *If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, then*

$$\mathbb{H}[X, Y] = \mathbb{H}[Y] + \mathbb{H}[X|Y].$$

Proof. Direct computation. □

Definition 14 (Mutual information). *If $X : \Omega \to S$, $Y : \Omega \to T$ are random variables, then*

$$\mathbb{I}[X : Y] := \mathbb{H}[X] + \mathbb{H}[Y] - \mathbb{H}[X, Y].$$

Lemma 15 (Alternative formulae for mutual information). *With notation as above, we have*

$$\mathbb{I}[X : Y] = \mathbb{I}[Y : X]$$

$$\mathbb{I}[X : Y] = \mathbb{H}[X] - \mathbb{H}[X|Y]$$

$$\mathbb{I}[X : Y] = \mathbb{H}[Y] - \mathbb{H}[Y|X]$$

Proof. Immediate from Lemmas 10, 13. □

Lemma 16 (Nonnegativity of mutual information). *We have $\mathbb{I}[X : Y] \geq 0$.*

Proof. An application of jensen's inequality and Theorems 15 and 49. □

Corollary 17 (Subadditivity). *With notation as above, we have $\mathbb{H}[X, Y] \leq \mathbb{H}[X] + \mathbb{H}[Y]$.*

Proof. Use Theorem 16. □

Corollary 18 (Conditioning reduces entropy). *With notation as above, we have $\mathbb{H}[X|Y] \leq \mathbb{H}[X]$.*

Proof. Combine Theorem 16 with Theorem 15. □

Corollary 19 (Submodularity). *With three random variables X, Y, Z, one has $\mathbb{H}[X|Y, Z] \leq \mathbb{H}[X|Z]$.*

Proof. Apply the "averaging over conditioning" argument to Theorem 18. □

Corollary 20 (Alternate form of submodularity). *With three random variables X, Y, Z, one has*

$$\mathbb{H}[X, Y, Z] + \mathbb{H}[Z] \leq \mathbb{H}[X, Z] + \mathbb{H}[Y, Z].$$

Proof. Apply Theorem 19 and Theorem 13. □

Lemma 21 (Vanishing of mutual information). *If X, Y are random variables, then $\mathbb{I}[X : Y] = 0$ if and only if X, Y are independent.*

Proof. An application of the equality case of Jensen's inequality and Theorem 49. □

Corollary 22 (Additivity of entropy). *If X, Y are random variables, then $\mathbb{H}[X, Y] = \mathbb{H}[X] + \mathbb{H}[Y]$ if and only if X, Y are independent.*

Proof. Direct from Theorem 21. □

Definition 23 (Conditional mutual information)**.** *If X, Y, Z are random variables, with Z U-valued, then*

$$\mathbb{I}[X:Y|Z] := \sum_{z \in U} P[Z=z]\mathbb{I}[(X|Z=z):(Y|Z=z)].$$

Lemma 24 (Alternate formula for conditional mutual information)**.** *We have*

$$\mathbb{I}[X:Y|Z] := \mathbb{H}[X|Z] + \mathbb{H}[Y|Z] - \mathbb{H}[X,Y|Z]$$

and

$$\mathbb{I}[X:Y|Z] := \mathbb{H}[X|Z] - \mathbb{H}[X|Y,Z].$$

Proof. Routine computation. □

Lemma 25 (Nonnegativity of conditional mutual information)**.** *If X, Y, Z are random variables, then $\mathbb{I}[X:Y|Z] \geq 0$.*

Proof. Use Theorem 23 and Theorem 19. □

Definition 26 (Conditionally independent random variables)**.** *Two random variables $X : \Omega \to S$ and $Y : \Omega \to T$ are conditionally independent relative to another random variable $Z : \Omega \to U$ if $P[X = s \wedge Y = t|Z = u] = P[X = s|Z = u]P[Y = t|Z = u]$ for all $s \in S, t \in T, u \in U$. (We won't need conditional independence for more variables than this.)*

Lemma 27 (Vanishing conditional mutual information)**.** *If X, Y, Z are random variables, then $\mathbb{I}[X:Y|Z] = 0$ iff X, Y are conditionally independent over Z.*

Proof. Immediate from Theorem 21 and Theorem 26. □

Corollary 28 (Entropy of conditionally independent variables)**.** *If X, Y are conditionally independent over Z, then*

$$\mathbb{H}[X,Y,Z] = \mathbb{H}[X,Z] + \mathbb{H}[Y,Z] - \mathbb{H}[Z].$$

Proof. Immediate from Theorem 27 and Theorem 24. □

3 Entropic Ruzsa calculus

In this section G will be a finite additive group. (May eventually want to generalize to infinite G.)

Lemma 29 (Negation preserves entropy)**.** *If X is G-valued, then $\mathbb{H}[-X] = \mathbb{H}[X]$.*

Proof. Immediate from Theorem 5. □

Lemma 30 (Shearing preserves entropy)**.** *If X, Y are G-valued, then $\mathbb{H}[X \pm Y|Y] = \mathbb{H}[X|Y]$ and $\mathbb{H}[X \pm Y, Y] = \mathbb{H}[X,Y]$.*

Proof. Immediate from Theorem 12 and Theorem 13. □

Lemma 31 (Lower bound of sumset)**.** *If X, Y are G-valued random variables on Ω, we have*

$$\max(\mathbb{H}[X], \mathbb{H}[Y]) - \mathbb{I}[X:Y] \leq \mathbb{H}[X \pm Y].$$

Proof. By Theorem 18, 30, 15, 29 we have

$$\mathbb{H}[X \pm Y] \geq \mathbb{H}[X \pm Y|Y] = \mathbb{H}[X|Y] = \mathbb{H}[X] - \mathbb{I}[X : Y]$$

and similarly with the roles of X, Y reversed, giving the claim. □

Corollary 32 (Independent lower bound on sumset)**.** *If X, Y are independent G-valued random variables, then*

$$\max(\mathbb{H}[X], \mathbb{H}[Y]) \leq \mathbb{H}[X \pm Y].$$

Proof. Combine Theorem 31 with Theorem 21. □

One random variable is said to be a copy of another if they have the same distribution.

Lemma 33 (Copy preserves entropy)**.** *If X' is a copy of X then $\mathbb{H}[X'] = \mathbb{H}[X]$.*

Proof. Immediate from Theorem 4. □

Lemma 34 (Existence of independent copies)**.** *Let $X_i : \Omega_i \to S_i$ be random variables for $i = 1, \ldots, k$. Then if one gives $\prod_{i=1}^k S_i$ the product measure of the laws of X_i, the coordinate functions $(x_j)_{j=1}^k \mapsto x_i$ are jointly independent random variables which are copies of the $X_1, \ldots, X_k$.*

Proof. Explicit computation. □

Definition 35 (Ruzsa distance)**.** *Let X, Y be G-valued random variables (not necessarily on the same sample space). The* Ruzsa distance *$d[X; Y]$ between X and Y is defined to be*

$$d[X; Y] := \mathbb{H}[X' - Y'] - \mathbb{H}[X']/2 - \mathbb{H}[Y']/2$$

where X', Y' are (the canonical) independent copies of X, Y from Theorem 34.

Lemma 36 (Copy preserves Ruzsa distance)**.** *If X', Y' are copies of X, Y respectively then $d[X'; Y'] = d[X; Y]$.*

Proof. Immediate from Definitions 35 and Theorem 33. □

Lemma 37 (Ruzsa distance in independent case)**.** *If X, Y are independent G-random variables then*

$$d[X; Y] := \mathbb{H}[X - Y] - \mathbb{H}[X]/2 - \mathbb{H}[Y]/2.$$

Proof. Immediate from Theorem 35 and Lemmas 5, 33. □

Lemma 38 (Distance controls entropy difference)**.** *If X, Y are G-valued random variables, then*

$$|\mathbb{H}[X] - H[Y]| \leq 2d[X; Y].$$

Proof. Immediate from Theorem 32 and Theorem 35, and also Theorem 29. □

Lemma 39 (Distance nonnegative)**.** *If X, Y are G-valued random variables, then*

$$d[X; Y] \geq 0.$$

Proof. Immediate from Theorem 38. □

Lemma 40 (Improved Ruzsa triangle inequality)**.** *If X, Y, Z are G-valued random variables on Ω with (X, Y) independent of Z, then*

$$\mathbb{H}[X - Y] \leq \mathbb{H}[X - Z] + \mathbb{H}[Z - Y] - \mathbb{H}[Z] \tag{3}$$

This is an improvement over the usual Ruzsa triangle inequality because X, Y are not assumed to be independent. However we will not utilize this improvement here.

Proof. Apply Theorem 20 to obtain

$$\mathbb{H}[X - Z, X - Y] + \mathbb{H}[Y, X - Y] \geq \mathbb{H}[X - Z, Y, X - Y] + \mathbb{H}[X - Y].$$

Using

$$\mathbb{H}[X - Z, X - Y] \leq \mathbb{H}[X - Z] + \mathbb{H}[Y - Z]$$

(from Theorem 5, Theorem 17),

$$\mathbb{H}[Y, X - Y] = \mathbb{H}[X, Y]$$

(from Theorem 5), and

$$\mathbb{H}[X - Z, Y, X - Y] = \mathbb{H}[X, Y, Z] = \mathbb{H}[X, Y] + \mathbb{H}[Z]$$

(from Theorem 5 and Theorem 22) and rearranging, we indeed obtain (3). □

Lemma 41 (Ruzsa triangle inequality)**.** *If X, Y, Z are G-valued random variables, then*

$$d[X; Y] \leq d[X; Z] + d[Z; Y].$$

Proof. By Theorem 36 and Lemmas 34, 37, it suffices to prove this inequality assuming that X, Y, Z are defined on the same space and are independent. But then the claim follows from Theorem 40 and Theorem 35. □

Definition 42 (Conditioned Ruzsa distance)**.** *If (X, Z) and (Y, W) are random variables (where X and Y are G-valued) we define*

$$d[X|Z; Y|W] := \sum_{z,w} \mathbb{P}[Z = z]\mathbb{P}[W = w] d[(X|Z = z); (Y|(W = w))].$$

similarly

$$d[X; Y|W] := \sum_{w} \mathbb{P}[W = w] d[X; (Y|(W = w))].$$

Lemma 43 (Alternate form of distance)**.** *The expression $d[X|Z; Y|W]$ is unchanged if (X, Z) or (Y, W) is replaced by a copy. Furthermore, if (X, Z) and (Y, W) are independent, then*

$$d[X|Z; Y|W] = \mathbb{H}[X - Y|Z, W] - \mathbb{H}[X|Z]/2 - \mathbb{H}[Y|W]/2$$

and similarly

$$d[X; Y|W] = \mathbb{H}[X - Y|W] - \mathbb{H}[X]/2 - \mathbb{H}[Y|W]/2.$$

Proof. Straightforward thanks to Theorem 33, Theorem 36, Theorem 37, Theorem 42, Theorem 11. □

Lemma 44 (Kaimanovich-Vershik-Madiman inequality)**.** *Suppose that X, Y, Z are independent G-valued random variables. Then*

$$\mathbb{H}[X+Y+Z] - \mathbb{H}[X+Y] \leq \mathbb{H}[Y+Z] - \mathbb{H}[Y].$$

Proof. From Theorem 19 we have

$$\mathbb{H}[X, X+Y+Z] + \mathbb{H}[Z, X+Y+Z] \geq \mathbb{H}[X, Z, X+Y+Z] + \mathbb{H}[X+Y+Z].$$

However, using Lemmas 22, 5 repeatedly we have $\mathbb{H}[X, X+Y+Z] = \mathbb{H}[X, Y+Z] = \mathbb{H}[X] + \mathbb{H}[Y+Z]$, $\mathbb{H}[Z, X+Y+Z] = \mathbb{H}[Z, X+Y] = \mathbb{H}[Z] + \mathbb{H}[X+Y]$ and $\mathbb{H}[X, Z, X+Y+Z] = \mathbb{H}[X, Y, Z] = \mathbb{H}[X] + \mathbb{H}[Y] + \mathbb{H}[Z]$. The claim then follows from a calculation. □

Lemma 45 (Existence of conditional independent trials)**.** *For X, Y random variables, there exist random variables X_1, X_2, Y' on a common probability space with $(X_1, Y'), (X_2, Y')$ both having the distribution of (X, Y), and X_1, X_2 conditionally independent over Y' in the sense of Theorem 26.*

Proof. Explicit construction. □

Lemma 46 (Balog-Szemerédi-Gowers)**.** *Let A, B be G-valued random variables on Ω, and set $Z := A + B$. Then*

$$\sum_z \mathbb{P}[Z = z] d[(A|Z=z); (B|Z=z)] \leq 3\mathbb{I}[A:B] + 2\mathbb{H}[Z] - \mathbb{H}[A] - \mathbb{H}[B]. \tag{4}$$

Proof. Let (A_1, B_1) and (A_2, B_2) (and Z', which by abuse of notation we call Z) be conditionally independent trials of (A, B) relative to Z as produced by Theorem 45, thus (A_1, B_1) and (A_2, B_2) are coupled through the random variable $A_1 + B_1 = A_2 + B_2$, which by abuse of notation we shall also call Z.

Observe from Theorem 37 that the left-hand side of (4) is

$$\mathbb{H}[A_1 - B_2|Z] - \mathbb{H}[A_1|Z]/2 - \mathbb{H}[B_2|Z]/2. \tag{5}$$

since, crucially, $(A_1|Z=z)$ and $(B_2|Z=z)$ are independent for all z.

Applying submodularity (Theorem 20) gives

$$\begin{aligned} &\mathbb{H}[A_1 - B_2] + \mathbb{H}[A_1 - B_2, A_1, B_1] \\ &\quad \leq \mathbb{H}[A_1 - B_2, A_1] + \mathbb{H}[A_1 - B_2, B_1]. \end{aligned} \tag{6}$$

We estimate the second, third and fourth terms appearing here. First note that, by Theorem 28 and Theorem 5 (noting that the tuple $(A_1 - B_2, A_1, B_1)$ determines the tuple (A_1, A_2, B_1, B_2) since $A_1 + B_1 = A_2 + B_2$)

$$\mathbb{H}[A_1 - B_2, A_1, B_1] = \mathbb{H}[A_1, B_1, A_2, B_2, Z] = 2\mathbb{H}[A, B] - \mathbb{H}[Z]. \tag{7}$$

Next observe that

$$\mathbb{H}[A_1 - B_2, A_1] = \mathbb{H}[A_1, B_2] \leq \mathbb{H}[A] + \mathbb{H}[B]. \tag{8}$$

Finally, we have

$$\mathbb{H}[A_1 - B_2, B_1] = \mathbb{H}[A_2 - B_1, B_1] = \mathbb{H}[A_2, B_1] \leq \mathbb{H}[A] + \mathbb{H}[B]. \tag{9}$$

Substituting (7), (8) and (9) into (6) yields

$$\mathbb{H}[A_1 - B_2] \leq 2\mathbb{I}[A:B] + \mathbb{H}[Z]$$

and so by Theorem 18

$$\mathbb{H}[A_1 - B_2|Z] \leq 2\mathbb{I}[A:B] + \mathbb{H}[Z].$$

Since

$$\begin{aligned}\mathbb{H}[A_1|Z] &= \mathbb{H}[A_1, A_1 + B_1] - \mathbb{H}[Z] \\ &= \mathbb{H}[A, B] - \mathbb{H}[Z] \\ &= \mathbb{H}[Z] - \mathbb{I}[A : B] - 2\mathbb{H}[Z] - \mathbb{H}[A] - \mathbb{H}[B]\end{aligned}$$

and similarly for $\mathbb{H}[B_2|Z]$, we see that (5) is bounded by $3\mathbb{I}[A : B] + 2\mathbb{H}[Z] - \mathbb{H}[A] - \mathbb{H}[B]$ as claimed. □

Lemma 47 (Upper bound on conditioned Ruzsa distance)**.** *Suppose that (X, Z) and (Y, W) are random variables, where X, Y take values in an abelian group. Then*

$$d[X|Z; Y|W] \leq d[X; Y] + \tfrac{1}{2}\mathbb{I}[X : Z] + \tfrac{1}{2}\mathbb{I}[Y : W].$$

In particular,

$$d[X; Y|W] \leq d[X; Y] + \tfrac{1}{2}\mathbb{I}[Y : W].$$

Proof. Using Theorem 43 and Theorem 34, if $(X', Z'), (Y', W')$ are independent copies of the variables (X, Z), (Y, W), we have

$$\begin{aligned}d[X|Z; Y|W] &= \mathbb{H}[X' - Y'|Z', W'] - \tfrac{1}{2}\mathbb{H}[X'|Z'] - \tfrac{1}{2}H[Y'|W'] \\ &\leq \mathbb{H}[X' - Y'] - \tfrac{1}{2}\mathbb{H}[X'|Z'] - \tfrac{1}{2}H[Y'|W'] \\ &= d[X'; Y'] + \tfrac{1}{2}\mathbb{I}[X' : Z'] + \tfrac{1}{2}\mathbb{I}[Y' : W'].\end{aligned}$$

Here, in the middle step we used Theorem 18, and in the last step we used Theorem 35 and Theorem 14. □

Lemma 48 (Comparison of Ruzsa distances, I)**.** *Let X, Y, Z be random variables taking values in some abelian group of characteristic 2, and with Y, Z independent. Then we have*

$$\begin{aligned}d[X; Y + Z] - d[X; Y] &\leq \tfrac{1}{2}(\mathbb{H}[Y + Z] - \mathbb{H}[Y]) \\ &= \tfrac{1}{2}d[Y; Z] + \tfrac{1}{4}\mathbb{H}[Z] - \tfrac{1}{4}\mathbb{H}[Y]. \qquad (10)\end{aligned}$$

and

$$\begin{aligned}d[X; Y|Y + Z] - d[X; Y] &\leq \tfrac{1}{2}\big(\mathbb{H}[Y + Z] - \mathbb{H}[Z]\big) \\ &= \tfrac{1}{2}d[Y; Z] + \tfrac{1}{4}\mathbb{H}[Y] - \tfrac{1}{4}\mathbb{H}[Z]. \qquad (11)\end{aligned}$$

Proof. We first prove (10). We may assume (taking an independent copy, using Theorem 34 and Theorem 36, 37) that X is independent of Y, Z. Then we have

$$\begin{aligned}&d[X; Y + Z] - d[X; Y] \\ &\quad = \mathbb{H}[X + Y + Z] - \mathbb{H}[X + Y] - \tfrac{1}{2}\mathbb{H}[Y + Z] + \tfrac{1}{2}\mathbb{H}[Y].\end{aligned}$$

Combining this with Theorem 44 gives the required bound. The second form of the result is immediate Theorem 37.

Turning to (11), we have from Theorem 14 and Theorem 5

$$\begin{aligned}\mathbb{I}[Y : Y + Z] &= \mathbb{H}[Y] + \mathbb{H}[Y + Z] - \mathbb{H}[Y, Y + Z] \\ &= \mathbb{H}[Y] + \mathbb{H}[Y + Z] - \mathbb{H}[Y, Z] = \mathbb{H}[Y + Z] - \mathbb{H}[Z],\end{aligned}$$

and so (11) is a consequence of Theorem 47. Once again the second form of the result is immediate from Theorem 37. □

4 Applications of Jensen's inequality

In this chapter, h denotes the function $h(x) := x \log \frac{1}{x}$ for $x \in [0,1]$.

Lemma 49 (Concavity). *h is strictly concave on $[0,\infty)$.*

Proof. Check that h' is strictly monotone decreasing. □

Lemma 50 (τ depends only on distribution). *If X_1', X_2' are copies of X_1, X_2, then $\tau[X_1'; X_2'] = \tau[X_1; X_2]$.*

Proof. Immediate from Theorem 33. □

Proposition 51 (τ has minimum). *A pair X_1, X_2 of τ-minimizers exist.*

Proof. By Theorem 50, τ only depends on the probability distributions of X_1, X_2. This ranges over a compact space, and τ is continuous. So τ has a minimum. □

5 Conclusion

Theorem 52 (Entropy version of PFR). *Let $G = \mathbb{F}_2^n$, and suppose that X_1^0, X_2^0 are G-valued random variables. Then there is some subgroup $H \leq G$ such that*

$$d[X_1^0; U_H] + d[X_2^0; U_H] \leq 11 d[X_1^0; X_2^0],$$

where U_H is uniformly distributed on H. Furthermore, both $d[X_1^0; U_H]$ and $d[X_2^0; U_H]$ are at most $6d[X_1^0; X_2^0]$.

6 The m-torsion case

7 Data processing inequality

Lemma 53 (Data processing for a single variable). *Let X be a random variable. Then for any function f on the range of X, one has $\mathbb{H}[f(X)] \leq \mathbb{H}[X]$.*

Proof. We have

$$\mathbb{H}[X] = \mathbb{H}[X, f(X)] = \mathbb{H}[f(X)] + \mathbb{H}[X|f(X)]$$

thanks to Theorem 5 and Theorem 13, giving the claim. □

Lemma 54 (One-sided unconditional data processing inequality). *Let X, Y be random variables. For any function f, g on the range of X, we have $\mathbb{I}[f(X) : Y] \leq \mathbb{I}[X : Y]$.*

Proof. By Theorem 15 it suffices to show that $\mathbb{H}[Y|X] \leq \mathbb{H}[Y|f(X)]$. But this follows from Theorem 19 (and Theorem 5). □

Lemma 55 (Unconditional data processing inequality). *Let X, Y be random variables. For any functions f, g on the ranges of X, Y respectively, we have $\mathbb{I}[f(X) : g(Y)] \leq \mathbb{I}[X : Y]$.*

Proof. From Theorem 54, Theorem 10 we have $\mathbb{I}[f(X) : Y] \leq \mathbb{I}[X : Y]$ and $\mathbb{I}[f(X) : g(Y)] \leq \mathbb{I}[f(X) : Y]$, giving the claim. □

Lemma 56 (Data processing inequality). *Let X, Y, Z. For any functions f, g on the ranges of X, Y respectively, we have $\mathbb{I}[f(X) : g(Y)|Z] \leq \mathbb{I}[X : Y|Z]$.*

Proof. Apply Theorem 55 to X, Y conditioned to the event $Z = z$, multiply by $\mathbf{P}[Z = z]$, and sum using Theorem 23. □

8 More Ruzsa distance estimates

Let G be an additive group.

Lemma 57 (Flipping a sign)**.** *If X, Y are G-valued, then*

$$d[X;-Y] \leq 3d[X;Y].$$

Proof. Without loss of generality (using Theorem 36 and Theorem 34) we may take X, Y to be independent. By (X_1, Y_1), (X_2, Y_2) be copies of (X, Y) that are conditionally independent over $X_1 - Y_1 = X_2 - Y_2$ (this exists thanks to Theorem 45). By Theorem 34, we can also find another copy (X_3, Y_3) of (X, Y) that is independent of X_1, Y_1, X_2, Y_2. From Theorem 20, one has

$$\mathbb{H}[X_3 - Y_2, X_1 - Y_3, X_2, Y_1, X_3, Y_3, X_3 + Y_3] + \mathbb{H}[X_3 + Y_3] \leq \mathbb{H}[X_3 - Y_2, X_1 - Y_3, X_2, Y_1, X_3 + Y_3] + \mathbb{H}[X_3, Y_3, X_3 + Y_3].$$

From Theorem 37, Theorem 29, Theorem 36 we have

$$\mathbb{H}[X_3 + Y_3] = \frac{1}{2}\mathbb{H}[X_3] + \frac{1}{2}\mathbb{H}[-Y_3] + d[X_3; -Y_3] = \frac{1}{2}\mathbb{H}[X] + \frac{1}{2}\mathbb{H}[Y] + d[X; -Y].$$

Since $X_3 + Y_3$ is a function of X_3, Y_3, we see from Theorem 5 and Theorem 22 that

$$\mathbb{H}[X_3, Y_3, X_3 + Y_3] = \mathbb{H}[X_3, Y_3] = \mathbb{H}[X, Y] = \mathbb{H}[X] + \mathbb{H}[Y].$$

Because $X_1 - Y_1 = X_2 - Y_2$, we have

$$X_3 + Y_3 = (X_3 - Y_2) - (X_1 - Y_3) + (X_2 + Y_1)$$

and thus by Theorem 5

$$\mathbb{H}[X_3 - Y_2, X_1 - Y_3, X_2, Y_1, X_3 + Y_3] = \mathbb{H}[X_3 - Y_2, X_1 - Y_3, X_2, Y_1]$$

and hence by Theorem 17

$$\mathbb{H}[X_3 - Y_2, X_1 - Y_3, X_2, Y_1, X_3 + Y_3] \leq \mathbb{H}[X_3 - Y_2] + \mathbb{H}[X_1 - Y_3] + \mathbb{H}[X_2] + \mathbb{H}[Y_1].$$

Since X_3, Y_2 are independent, we see from Theorem 37, Theorem 36 that

$$\mathbb{H}[X_3 - Y_2] = \frac{1}{2}\mathbb{H}[X] + \frac{1}{2}\mathbb{H}[Y] + d[X;Y].$$

Similarly

$$\mathbb{H}[X_1 - Y_3] = \frac{1}{2}\mathbb{H}[X] + \frac{1}{2}\mathbb{H}[Y] + d[X;Y].$$

We conclude that

$$\mathbb{H}[X_3 - Y_2, X_1 - Y_3, X_2, Y_1, X_3 + Y_3] \leq 2\mathbb{H}[X] + 2\mathbb{H}[Y] + 2d[X;Y].$$

Finally, from Theorem 53 we have

$$\mathbb{H}[X_1, Y_1, X_2, Y_2, X_3, Y_3] \leq \mathbb{H}[X_3 - Y_2, X_1 - Y_3, X_2, Y_1, X_3, Y_3, X_3 + Y_3].$$

From Theorem 22 followed by Theorem 28, we have

$$\mathbb{H}[X_1, Y_1, X_2, Y_2, X_3, Y_3] = \mathbb{H}[X_1, Y_1, X_1 - Y_1] + \mathbb{H}[X_2, Y_2, X_2 - Y_2] - \mathbb{H}[X_1 - Y_1] + \mathbb{H}[X_3, Y_3]$$

and thus by Theorem 37, Theorem 36, Theorem 5, Theorem 22

$$\mathbb{H}[X_1, Y_1, X_2, Y_2, X_3, Y_3] = \mathbb{H}[X] + \mathbb{H}[Y] + \mathbb{H}[X] + \mathbb{H}[Y] - \left(\frac{1}{2}\mathbb{H}[X] + \frac{1}{2}\mathbb{H}[Y] + d[X;Y]\right) + \mathbb{H}[X] + \mathbb{H}[Y].$$

Applying all of these estimates, the claim now follows from linear arithmetic. □

Lemma 58 (Kaimonovich–Vershik–Madiman inequality)**.** *If $n \geq 0$ and $X, Y_1, \ldots, Y_n$ are jointly independent G-valued random variables, then*

$$\mathbb{H}\left[X + \sum_{i=1}^n Y_i\right] - \mathbb{H}[X] \leq \sum_{i=1}^n \left(\mathbb{H}[X + Y_i] - \mathbb{H}[X]\right).$$

Proof. This is trivial for $n = 0, 1$, while the $n = 2$ case is Theorem 44. Now suppose inductively that $n > 2$, and the claim was already proven for $n - 1$. By a further application of Theorem 44 one has

$$\mathbb{H}\left[X + \sum_{i=1}^n Y_i\right] - \mathbb{H}\left[X + \sum_{i=1}^{n-1} Y_i\right] \leq \mathbb{H}[X + Y_n] - \mathbb{H}[X].$$

By induction hypothesis one has

$$\mathbb{H}\left[X + \sum_{i=1}^{n-1} Y_i\right] - \mathbb{H}[X] \leq \sum_{i=1}^{n-1} \mathbb{H}[X + Y_i] - \mathbb{H}[X].$$

Summing the two inequalities, we obtain the claim. □

Lemma 59 (Kaimonovich–Vershik–Madiman inequality, II)**.** *If $n \geq 1$ and $X, Y_1, \ldots, Y_n$ are jointly independent G-valued random variables, then*

$$d[X; \sum_{i=1}^n Y_i] \leq 2 \sum_{i=1}^n d[X; Y_i].$$

Proof. Applying Theorem 58 with all the Y_i replaced by $-Y_i$, and using Theorem 29 and Theorem 37, we obtain after some rearranging

$$d[X; \sum_{i=1}^n Y_i] + \frac{1}{2}(\mathbb{H}[\sum_{i=1}^n Y_i] - \mathbb{H}[X]) \leq \sum_{i=1}^n \left(d[X; Y_i] + \frac{1}{2}(\mathbb{H}[Y_i] - \mathbb{H}[X])\right).$$

From Theorem 32 we have

$$\mathbb{H}[\sum_{i=1}^n Y_i] \geq \mathbb{H}[Y_i]$$

for all i; subtracting $\mathbb{H}[X]$ and averaging, we conclude that

$$\mathbb{H}[\sum_{i=1}^n Y_i] - \mathbb{H}[X] \geq \frac{1}{n}\sum_{i=1}^n \mathbb{H}[Y_i] - \mathbb{H}[X]$$

and thus

$$d[X; \sum_{i=1}^n Y_i] \leq \sum_{i=1}^n d[X; Y_i] + \frac{n-1}{2n}(\mathbb{H}[Y_i] - \mathbb{H}[X]).$$

From Theorem 38 we have

$$\mathbb{H}[Y_i] - \mathbb{H}[X] \leq 2d[X;Y_i].$$

Since $0 \leq \frac{n-1}{2n} \leq \frac{1}{2}$, the claim follows. □

Lemma 60 (Kaimonovich–Vershik–Madiman inequality, III). *If $n \geq 1$ and $X, Y_1, \ldots, Y_n$ are jointly independent G-valued random variables, then*

$$d\left[X; \sum_{i=1}^n Y_i\right] \leq d\left[X; Y_1\right] + \frac{1}{2}\left(\mathbb{H}\left[\sum_{i=1}^n Y_i\right] - \mathbb{H}[Y_1]\right).$$

Proof. From Theorem 44 one has

$$\mathbb{H}\left[-X + \sum_{i=1}^n Y_i\right] \leq \mathbb{H}[-X + Y_1] + \mathbb{H}\left[\sum_{i=1}^n Y_i\right] - \mathbb{H}[Y_1].$$

The claim then follows from Theorem 37 and some elementary algebra. □

Lemma 61 (Comparing sums). *Let $(X_i)_{1\leq i\leq m}$ and $(Y_j)_{1\leq j\leq l}$ be tuples of jointly independent random variables (so the X's and Y's are also independent of each other), and let $f : \{1, \ldots, l\} \to \{1, \ldots, m\}$ be a function, then*

$$\mathbb{H}[\sum_{j=1}^l Y_j] \leq \mathbb{H}[\sum_{i=1}^m X_i] + \sum_{j=1}^l (\mathbb{H}[Y_j - X_{f(j)}] - \mathbb{H}[X_{f(j)}]).$$

Proof. Write $W := \sum_{i=1}^m X_i$. From Theorem 32 we have

$$\mathbb{H}[\sum_{j=1}^l Y_j] \leq \mathbb{H}[-W + \sum_{j=1}^l Y_j]$$

while from Theorem 58 one has

$$\mathbb{H}[-W + \sum_{j=1}^l Y_j] \leq \mathbb{H}[-W] + \sum_{j=1}^l \mathbb{H}[-W + Y_j] - \mathbb{H}[-W].$$

From Theorem 44 one has

$$\mathbb{H}[-W + Y_j] - \mathbb{H}[-W] \leq \mathbb{H}[-X_{f(j)} + Y_j] - \mathbb{H}[-X_{f(j)}].$$

The claim now follows from Theorem 29 and some elementary algebra. □

Lemma 62 (Sums of dilates I). *Let X, Y, X' be independent G-valued random variables, with X' a copy of X, and let a be an integer. Then*

$$\mathbb{H}[X - (a+1)Y] \leq \mathbb{H}[X - aY] + \mathbb{H}[X - Y - X'] - \mathbb{H}[X]$$

and

$$\mathbb{H}[X - (a-1)Y] \leq \mathbb{H}[X - aY] + \mathbb{H}[X - Y - X'] - \mathbb{H}[X].$$

Proof. From Theorem 40 we have

$$\mathbb{H}[(X-Y)-aY] \leq \mathbb{H}[(X-Y)-X'] + \mathbb{H}[X'-aY] - \mathbb{H}[X']$$

which gives the first inequality. Similarly from Theorem 40 we have

$$\mathbb{H}[(X+Y)-aY] \leq \mathbb{H}[(X+Y)-X'] + \mathbb{H}[X'-aY] - \mathbb{H}[X']$$

which (when combined with Theorem 29) gives the second inequality. □

Lemma 63 (Sums of dilates II)**.** *Let X, Y be independent G-valued random variables, and let a be an integer. Then*

$$\mathbb{H}[X-aY] - \mathbb{H}[X] \leq 4|a| d[X;Y].$$

Proof. From Theorem 44 one has

$$\mathbb{H}[Y-X+X'] - \mathbb{H}[Y-X] \leq \mathbb{H}[Y+X'] - \mathbb{H}[Y] = \mathbb{H}[Y+X] - \mathbb{H}[Y]$$

which by Theorem 37 gives

$$\mathbb{H}[X-Y-X'] - \mathbb{H}[X] \leq d[X;Y] + d[X;-Y]$$

and hence by Theorem 57

$$\mathbb{H}[X-Y-X'] - \mathbb{H}[X] \leq 4d[X;Y].$$

From Theorem 62 we then have

$$\mathbb{H}[X-(a\pm 1)Y] \leq \mathbb{H}[X-aY] + 4d[X;Y]$$

and the claim now follows by an induction on $|a|$. □

We remark that in the paper [GGMT2024] the variant estimate

$$\mathbb{H}[X-aY] - \mathbb{H}[X] \leq (4 + 10\lfloor \log_2 |a| \rfloor) d[X;Y]$$

is also proven by a similar method. This variant is superior for $|a| \geq 9$ (or $|a| = 7$); but we will not need this estimate here.

9 Multidistance

We continue to let G be an abelian group.

Definition 64 (Multidistance)**.** *Let m be a positive integer, and let $X_{[m]} = (X_i)_{1\leq i\leq m}$ be an m-tuple of G-valued random variables X_i. Then we define*

$$D[X_{[m]}] := \mathbb{H}[\sum_{i=1}^m \tilde{X}_i] - \frac{1}{m}\sum_{i=1}^m \mathbb{H}[\tilde{X}_i],$$

where the $\tilde{X}_i$ are independent copies of the X_i.

Lemma 65 (Multidistance of copy)**.** *If $X_{[m]} = (X_i)_{1\leq i\leq m}$ and $Y_{[m]} = (Y_i)_{1\leq i\leq m}$ are such that X_i and Y_i have the same distribution for each i, then $D[X_{[m]}] = D[Y_{[m]}]$.*

Proof. Clear from Lemma 33. □

Lemma 66 (Multidistance of independent variables). *If $X_{[m]} = (X_i)_{1\leq i\leq m}$ are jointly independent, then $D[X_{[m]}] = \mathbb{H}[\sum_{i=1}^m X_i] - \frac{1}{m}\sum_{i=1}^m \mathbb{H}[X_i]$.*

Proof. Clear from definition. □

Lemma 67 (Nonnegativity). *For any such tuple, we have $D[X_{[m]}] \geq 0$.*

Proof. From Theorem 32 one has

$$\mathbb{H}[\sum_{i=1}^m \tilde{X}_i] \geq \mathbb{H}[\tilde{X}_i]$$

for each $1 \leq i \leq m$. Averaging over i, we obtain the claim. □

Lemma 68 (Relabeling). *If $\phi : \{1,\dots,m\} \to \{1,\dots,m\}$ is a bijection, then $D[X_{[m]}] = D[(X_{\phi(j)})_{1\leq j\leq m}]$.*

Proof. Trivial. □

Lemma 69 (Multidistance and Ruzsa distance, I). *Let $m \geq 2$, and let $X_{[m]}$ be a tuple of G-valued random variables. Then*

$$\sum_{1\leq j,k\leq m : j\neq k} d[X_j; -X_k] \leq m(m-1)D[X_{[m]}].$$

Proof. By Theorem 36, Theorem 65 we may take the X_i to be jointly independent. From Theorem 32, we see that for any distinct $1 \leq j,k \leq m$, we have

$$\mathbb{H}[X_j + X_k] \leq \mathbb{H}[\sum_{i=1}^m X_i],$$

and hence by Theorem 37

$$d[X_j; -X_k] \leq \mathbb{H}[\sum_{i=1}^m X_i] - \frac{1}{2}\mathbb{H}[X_j] - \frac{1}{2}\mathbb{H}[X_k].$$

Summing this over all pairs (j,k), $j \neq k$ and using Lemma 66, we obtain the claim. □

Lemma 70 (Multidistance and Ruzsa distance, II). *Let $m \geq 2$, and let $X_{[m]}$ be a tuple of G-valued random variables. Then*

$$\sum_{j=1}^m d[X_j; X_j] \leq 2mD[X_{[m]}].$$

Proof. From Theorem 41 we have $d[X_j; X_j] \leq 2d[X_j; -X_k]$, and applying this to every summand in Theorem 69, we obtain the claim. □

Lemma 71 (Multidistance and Ruzsa distance, III). *Let $m \geq 2$, and let $X_{[m]}$ be a tuple of G-valued random variables. If the X_i all have the same distribution, then $D[X_{[m]}] \leq md[X_i; X_i]$ for any $1 \leq i \leq m$.*

Proof. By Theorem 36, Theorem 65 we may take the X_i to be jointly independent. Let X_0 be a further independent copy of the X_i. From Theorem 58, we have

$$\mathbb{H}[-X_0 + \sum_{i=1}^m X_i] - \mathbb{H}[-X_0] \leq \sum_{i=1}^m \mathbb{H}[X_0 - X_i] - \mathbb{H}[-X_0]$$

and hence by Theorem 29 and Theorem 37

$$\mathbb{H}[-X_0 + \sum_{i=1}^m X_i] - \mathbb{H}[X_0] \leq m d[X_i, X_i].$$

On the other hand, by Theorem 32 we have

$$\mathbb{H}[\sum_{i=1}^m X_i] \leq \mathbb{H}[-X_0 + \sum_{i=1}^m X_i]$$

and the claim follows. □

Lemma 72 (Multidistance and Ruzsa distance, IV)**.** *Let $m \geq 2$, and let $X_{[m]}$ be a tuple of independent G-valued random variables. Let $W := \sum_{i=1}^m X_i$. Then*

$$d[W; -W] \leq 2D[X_i].$$

Proof. Take $(X_i')_{1 \leq i \leq m}$ to be further independent copies of $(X_i)_{1 \leq i \leq m}$ (which exist by Theorem 34), and write $W' := \sum_{i=1}^m X_i'$. Fix any distinct $a, b \in I$.

From Theorem 44 one has

$$\mathbb{H}[W + W'] \leq \mathbb{H}[W] + \mathbb{H}[X_a + W'] - \mathbb{H}[X_a] \tag{12}$$

and also

$$\mathbb{H}[X_a + W'] \leq \mathbb{H}[X_a + X_b] + \mathbb{H}[W'] - \mathbb{H}[X_b'].$$

Combining this with (12) and then applying Theorem 32 we have

$$\begin{aligned} \mathbb{H}[W + W'] &\leq 2\mathbb{H}[W] + \mathbb{H}[X_a + X_b] - \mathbb{H}[X_a] - \mathbb{H}[X_b] \\ &\leq 3\mathbb{H}[W] - \mathbb{H}[X_a] - \mathbb{H}[X_b]. \end{aligned}$$

Averaging this over all choices of (a, b) gives $\mathbb{H}[W] + 2D[X_{[m]}]$, and the claim follows from Theorem 37. □

Proposition 73 (Vanishing)**.** *If $D[X_{[m]}] = 0$, then for each $1 \leq i \leq m$ there is a finite subgroup $H_i \leq G$ such that $d[X_i; U_{H_i}] = 0$.*

Proof. From Theorem 70 and Theorem 39 we have $d[X_j; X_j] = 0$ for all $1 \leq j \leq m$. The claim now follows from Theorem 103. □

With more effort one can show that H_i is independent of i, but we will not need to do so here.

10 The tau functional

Fix $m \geq 2$, and a reference variable X^0 in G.

Definition 74 (η)**.** *We set* $\eta := \frac{1}{32m^3}$.

Definition 75 (τ-functional)**.** *If* $(X_i)_{1\leq i\leq m}$ *is a tuple, we define its* τ*-functional*

$$\tau[(X_i)_{1\leq i\leq m}] := D[(X_i)_{1\leq i\leq m}] + \eta \sum_{i=1}^m d[X_i; X^0].$$

Definition 76 (τ-minimizer)**.** *A* τ*-minimizer is a tuple* $(X_i)_{1\leq i\leq m}$ *that minimizes the* τ*-functional among all tuples of* G*-valued random variables.*

Proposition 77 (Existence of τ-minimizer)**.** *If* G *is finite, then a* τ*-minimizer exists.*

Proof. This is similar to the proof of Theorem 51. □

Proposition 78 (Minimizer close to reference variables)**.** *If* $(X_i)_{1\leq i\leq m}$ *is a* τ*-minimizer, then* $\sum_{i=1}^m d[X_i; X^0] \leq \frac{2m}{\eta} d[X^0; X^0]$.

Proof. By Theorem 76 we have

$$\tau[(X_i)_{1\leq i\leq m}] \leq \tau[(X^0)_{1\leq i\leq m}]$$

and hence by Theorem 75 and Theorem 67

$$\eta \sum_{i=1}^m d[X_i; X^0] \leq D[(X^0)_{1\leq i\leq m}] + m d[X^0; X^0].$$

The claim now follows from Theorem 71. □

Lemma 79 (Lower bound on multidistance)**.** *If* $(X_i)_{1\leq i\leq m}$ *is a* τ*-minimizer, and* $k := D[(X_i)_{1\leq i\leq m}]$*, then for any other tuple* $(X'_i)_{1\leq i\leq m}$*, one has*

$$k - D[(X'_i)_{1\leq i\leq m}] \leq \eta \sum_{i=1}^m d[X_i; X'_i].$$

Proof. By Theorem 76 we have

$$\tau[(X_i)_{1\leq i\leq m}] \leq \tau[(X'_i)_{1\leq i\leq m}]$$

and hence by Theorem 75

$$k + \eta \sum_{i=1}^m d[X_i; X^0] \leq D[(X'_i)_{1\leq i\leq m}] + \eta \sum_{i=1}^m d[X'_i; X^0].$$

On the other hand, by Theorem 41 we have

$$d[X'_i; X^0] \leq d[X_i; X^0] + d[X_i; X'_i].$$

The claim follows. □

Definition 80 (Conditional multidistance)**.** *If $X_{[m]} = (X_i)_{1\le i\le m}$ and $Y_{[m]} = (Y_i)_{1\le i\le m}$ are tuples of random variables, with the X_i being G-valued (but the Y_i need not be), then we define*

$$D[X_{[m]}|Y_{[m]}] = \sum_{(y_i)_{1\le i\le m}} \Big(\prod_{1\le i\le m} p_{Y_i}(y_i)\Big) D[(X_i \,|\, Y_i = y_i)_{1\le i\le m}] \tag{13}$$

where each y_i ranges over the support of p_{Y_i} for $1 \le i \le m$.

Lemma 81 (Alternate form of conditional multidistance)**.** *If the (X_i, Y_i) are independent,*

$$D[X_{[m]}|Y_{[m]}] := \mathbb{H}[\sum_{i=1}^m X_i \big| (Y_j)_{1\le j\le m}] - \frac{1}{m}\sum_{i=1}^m \mathbb{H}[X_i|Y_i]. \tag{14}$$

Proof. This is routine from Theorem 11 and Definitions 64 and 80. □

Lemma 82 (Lower bound on conditional multidistance)**.** *If $(X_i)_{1\le i\le m}$ is a τ-minimizer, and $k := D[(X_i)_{1\le i\le m}]$, then for any other tuples $(X_i')_{1\le i\le m}$ and $(Y_i)_{1\le i\le m}$ with the X_i' G-valued, one has*

$$k - D[(X_i')_{1\le i\le m}|(Y_i)_{1\le i\le m}] \le \eta \sum_{i=1}^m d[X_i; X_i'|Y_i].$$

Proof. Immediate from Theorem 79, Theorem 81, and Theorem 42. □

Corollary 83 (Lower bound on conditional multidistance, II)**.** *With the notation of the previous lemma, we have*

$$k - D[X_{[m]}'|Y_{[m]}] \le \eta \sum_{i=1}^m d[X_{\sigma(i)}; X_i'|Y_i] \tag{15}$$

for any permutation $\sigma : \{1, \dots, m\} \to \{1, \dots, m\}$.

Proof. This follows from Theorem 82 and Theorem 68. □

11 The multidistance chain rule

Lemma 84 (Multidistance chain rule)**.** *Let $\pi\colon G \to H$ be a homomorphism of abelian groups and let $X_{[m]}$ be a tuple of jointly independent G-valued random variables. Then $D[X_{[m]}]$ is equal to*

$$D[X_{[m]}|\pi(X_{[m]})] + D[\pi(X_{[m]})] + \mathbb{I}[\sum_{i=1}^m X_i : \pi(X_{[m]}) \;\big|\; \pi\big(\sum_{i=1}^m X_i\big)] \tag{16}$$

where $\pi(X_{[m]}) := (\pi(X_i))_{1\le i\le m}$.

Proof. For notational brevity during this proof, write $S := \sum_{i=1}^m X_i$.

From Theorem 24 and Theorem 5, noting that $\pi(S)$ is determined both by S and by $\pi(X_{[m]})$, we have

$$\mathbb{I}[S : \pi(X_{[m]})|\pi(S)] = \mathbb{H}[S] + \mathbb{H}[\pi(X_{[m]})] - \mathbb{H}[S, \pi(X_{[m]})] - \mathbb{H}[\pi(S)],$$

and by Theorem 13 the right-hand side is equal to

$$\mathbb{H}[S] - \mathbb{H}[S|\pi(X_{[m]})] - \mathbb{H}[\pi(S)].$$

Therefore,

$$\mathbb{H}[S] = \mathbb{H}[S|\pi(X_{[m]})] + \mathbb{H}[\pi(S)] + \mathbb{I}[S : \pi(X_{[m]})|\pi(S)]. \tag{17}$$

From a further application of Theorem 13 and Theorem 5 we have

$$\mathbb{H}[X_i] = \mathbb{H}[X_i \,|\, \pi(X_i)] + \mathbb{H}[\pi(X_i)] \tag{18}$$

for all $1 \leq i \leq m$. Averaging (18) in i and subtracting this from (17), we obtain the claim from Theorem 64. □

We will need to iterate the multidistance chain rule, so it is convenient to observe a conditional version of this rule, as follows.

Lemma 85 (Conditional multidistance chain rule)**.** *Let $\pi\colon G \to H$ be a homomorphism of abelian groups. Let I be a finite index set and let $X_{[m]}$ be a tuple of G-valued random variables. Let $Y_{[m]}$ be another tuple of random variables (not necessarily G-valued). Suppose that the pairs (X_i, Y_i) are jointly independent of one another (but X_i need not be independent of Y_i). Then*

$$\begin{aligned} D[X_{[m]}|Y_{[m]}] = D[X_{[m]} \,|\, \pi(X_{[m]}), Y_{[m]}] + D[\pi(X_{[m]}) \,|\, Y_{[m]}] \\ + \mathbb{I}[\sum_{i=1}^m X_i : \pi(X_{[m]}) \;\big|\; \pi\big(\sum_{i=1}^m X_i\big), Y_{[m]}]. \end{aligned} \tag{19}$$

Proof. For each y_i in the support of p_{Y_i}, apply Theorem 84 with X_i replaced by the conditioned random variable $(X_i|Y_i = y_i)$, and the claim (19) follows by averaging (16) in the y_i using the weights p_{Y_i}. □

We can iterate the above lemma as follows.

Lemma 86. *Let m be a positive integer. Suppose one has a sequence*

$$G_m \to G_{m-1} \to \cdots \to G_1 \to G_0 = \{0\} \tag{20}$$

of homomorphisms between abelian groups $G_0, \ldots, G_m$, and for each $d = 0, \ldots, m$, let $\pi_d : G_m \to G_d$ be the homomorphism from G_m to G_d arising from this sequence by composition (so for instance π_m is the identity homomorphism and π_0 is the zero homomorphism). Let $X_{[m]} = (X_i)_{1 \leq i \leq m}$ be a jointly independent tuple of G_m-valued random variables. Then

$$\begin{aligned} D[X_{[m]}] = \sum_{d=1}^m D[\pi_d(X_{[m]}) \,|\, \pi_{d-1}(X_{[m]})] \\ + \sum_{d=1}^{m-1} \mathbb{I}[\sum_i X_i : \pi_d(X_{[m]}) \;\big|\; \pi_d\big(\sum_i X_i\big), \pi_{d-1}(X_{[m]})]. \end{aligned} \tag{21}$$

In particular, by Theorem 25,

$$\begin{aligned} D[X_{[m]}] \geq \sum_{d=1}^m D[\pi_d(X_{[m]})|\pi_{d-1}(X_{[m]})] \\ + \mathbb{I}[\sum_i X_i : \pi_1(X_{[m]}) \;\big|\; \pi_1\big(\sum_i X_i\big)]. \end{aligned} \tag{22}$$

Proof. From Theorem 85 (taking $Y_{[m]} = \pi_{d-1}(X_{[m]})$ and $\pi = \pi_d$ there, and noting that $\pi_d(X_{[m]})$ determines $Y_{[m]}$) we have

$$D[X_{[m]} \,|\, \pi_{d-1}(X_{[m]})] = D[X_{[m]} \,|\, \pi_d(X_{[m]})] + D[\pi_d(X_{[m]}) \,|\, \pi_{d-1}(X_{[m]})] + \mathbb{I}[\sum_{i=1}^m X_i : \pi_d(X_{[m]}) \;\big|\; \pi_d(\sum_{i=1}^m X_i), \pi_{d-1}(X_{[m]})]$$

for $d = 1, \dots, m$. The claim follows by telescoping series, noting that $D[X_{[m]}|\pi_0(X_{[m]})] = D[X_{[m]}]$ and that $\pi_m(X_{[m]}) = X_{[m]}$ (and also $\pi_m(\sum_i X_i) = \sum_i X_i$). □

In our application we will need the following special case of the above lemma.

Corollary 87. *Let G be an abelian group and let $m \geq 2$. Suppose that $X_{i,j}$, $1 \leq i, j \leq m$, are independent G-valued random variables. Then*

$$\begin{aligned}\mathbb{I}[(\sum_{i=1}^m X_{i,j})_{j=1}^m : (\sum_{j=1}^m X_{i,j})_{i=1}^m \;\big|\; \sum_{i=1}^m \sum_{j=1}^m X_{i,j}] \\ \leq \sum_{j=1}^{m-1} \Big(D[(X_{i,j})_{i=1}^m] - D[(X_{i,j})_{i=1}^m \;\big|\; (X_{i,j} + \cdots + X_{i,m})_{i=1}^m] \Big) \\ + D[(X_{i,m})_{i=1}^m] - D[(\sum_{j=1}^m X_{i,j})_{i=1}^m],\end{aligned}$$

where all the multidistances here involve the indexing set $\{1, \dots, m\}$.

Proof. In Theorem 86 we take $G_d := G^d$ with the maps $\pi_d \colon G^m \to G^d$ for $d = 1, \dots, m$ defined by

$$\pi_d(x_1, \dots, x_m) := (x_1, \dots, x_{d-1}, x_d + \cdots + x_m)$$

with $\pi_0 = 0$. Since $\pi_{d-1}(x)$ can be obtained from $\pi_d(x)$ by applying a homomorphism, we obtain a sequence of the form (20).

Now we apply Theorem 86 with $I = \{1, \dots, m\}$ and $X_i := (X_{i,j})_{j=1}^m$. Using joint independence and Theorem 22, we find that

$$D[X_{[m]}] = \sum_{j=1}^m D[(X_{i,j})_{1 \leq i \leq m}].$$

On the other hand, for $1 \leq j \leq m-1$, we see that once $\pi_j(X_i)$ is fixed, $\pi_{j+1}(X_i)$ is determined by $X_{i,j}$ and vice versa, so

$$D[\pi_{j+1}(X_{[m]}) \mid \pi_j(X_{[m]})] = D[(X_{i,j})_{1 \leq i \leq m} \mid \pi_j(X_{[m]})].$$

Since the $X_{i,j}$ are jointly independent, we may further simplify:

$$D[(X_{i,j})_{1 \leq i \leq m} \mid \pi_j(X_{[m]})] = D[(X_{i,j})_{1 \leq i \leq m} \mid (X_{i,j} + \cdots + X_{i,m})_{1 \leq i \leq m}].$$

Putting all this into the conclusion of Theorem 86, we obtain

$$\begin{aligned}\sum_{j=1}^m D[(X_{i,j})_{1\leq i\leq m}] \geq & \sum_{j=1}^{m-1} D[(X_{i,j})_{1\leq i\leq m} \mid (X_{i,j}+\cdots+X_{i,m})_{1\leq i\leq m}] \\ & + D[(\sum_{j=1}^m X_{i,j})_{1\leq i\leq m}] \\ & + \mathbb{I}[(\sum_{i=1}^m X_{i,j})_{j=1}^m : (\sum_{j=1}^m X_{i,j})_{i=1}^m \mid \sum_{i=1}^m \sum_{j=1}^m X_{i,j}]\end{aligned}$$

and the claim follows by rearranging. □

12 Bounding the mutual information

As before, G is an abelian group, and $m \geq 2$. We let $X_{[m]} = (X_i)_{i=1}^m$ be a τ-minimizer.

Proposition 88 (Bounding mutual information)**.** *Suppose that $X_{i,j}$, $1 \leq i,j \leq m$, are jointly independent G-valued random variables, such that for each $j = 1,\dots,m$, the random variables $(X_{i,j})_{i=1}^m$ coincide in distribution with some permutation of $X_{[m]}$. Write*

$$\mathcal{I} := \mathbb{I}[(\sum_{i=1}^m X_{i,j})_{j=1}^m : (\sum_{j=1}^m X_{i,j})_{i=1}^m \mid \sum_{i=1}^m \sum_{j=1}^m X_{i,j}].$$

Then

$$\mathcal{I} \leq m(4m+1)\eta k. \tag{23}$$

Proof. For each $j \in \{1,\dots,m\}$ we call the tuple $(X_{i,j})_{i=1}^m$ a *column* and for each $i \in \{1,\dots,m\}$ we call the tuple $(X_{i,j})_{j=1}^m$ a *row*. Hence, by hypothesis, each column is a permutation of $X_{[m]} = (X_i)_{i=1}^m$.

From Theorem 87 we have

$$\mathcal{I} \leq \sum_{j=1}^{m-1} A_j + B, \tag{24}$$

where

$$A_j := D[(X_{i,j})_{i=1}^m] - D[(X_{i,j})_{i=1}^m \mid (X_{i,j}+\cdots+X_{i,m})_{i=1}^m]$$

and

$$B := D[(X_{i,m})_{i=1}^m] - D[(\sum_{j=1}^m X_{i,j})_{i=1}^m].$$

We first consider the A_j, for fixed $j \in \{1,\dots,m-1\}$. By Theorem 68 and our hypothesis on columns, we have

$$D[(X_{i,j})_{i=1}^m] = D[(X_i)_{i=1}^m] = k.$$

Let $\sigma = \sigma_j \colon I \to I$ be a permutation such that $X_{i,j} \equiv X_{\sigma(i)}$, and write $X'_i := X_{i,j}$ and $Y_i := X_{i,j} + \cdots + X_{i,m}$. By Theorem 83, we have

$$A_j \le \eta(\sum_{i=1}^{m} d[X_{i,j}; X_{i,j}|X_{i,j} + \cdots + X_{i,m}]). \tag{25}$$

We similarly consider B. By Theorem 68 applied to the m-th column,

$$D[(X_{i,m})_{i=1}^m] = D[X_{[m]}] = k.$$

For $1 \le i \le m$, denote the sum of row i by

$$V_i := \sum_{j=1}^{m} X_{i,j};$$

if we apply Theorem 83 again, now with $X_{\sigma(i)} = X_{i,m}$, $X'_i := V_i$, and with the variable Y_i being trivial, we obtain

$$B \le \eta \sum_{i=1}^{m} d[X_{i,m}; V_i]. \tag{26}$$

It remains to bound the distances appearing in (25) and (26) further using Ruzsa calculus. For $1 \le j \le m-1$ and $1 \le i \le m$, by Theorem 48 we have

$$\begin{aligned} d[X_{i,j}; X_{i,j}|X_{i,j} + \cdots + X_{i,m}] &\le d[X_{i,j}; X_{i,j}] \\ &+ \tfrac{1}{2}\big(\mathbb{H}[X_{i,j} + \cdots + X_{i,m}] - \mathbb{H}[X_{i,j+1} + \cdots + X_{i,m}]\big). \end{aligned}$$

For each i, summing over $j = 1, \dots, m-1$ gives

$$\begin{aligned} &\sum_{j=1}^{m-1} d[X_{i,j}; X_{i,j}|X_{i,j} + \cdots + X_{i,m}] \\ &\qquad \le \sum_{j=1}^{m-1} d[X_{i,j}; X_{i,j}] + \frac{1}{2}\big(\mathbb{H}[V_i] - \mathbb{H}[X_{i,m}]\big). \end{aligned} \tag{27}$$

On the other hand, by Theorem 60 (since $X_{i,m}$ appears in the sum V_i) we have

$$d[X_{i,m}; V_i] \le d[X_{i,m}; X_{i,m}] + \frac{1}{2}\big(\mathbb{H}[V_i] - \mathbb{H}[X_{i,m}]\big). \tag{28}$$

Combining (24), (25) and (26) with (27) and (28) (the latter two summed over i), we get

$$\begin{aligned} \frac{1}{\eta}\mathcal{I} &\le \sum_{i,j=1}^{m} d[X_{i,j}; X_{i,j}] + \sum_{i=1}^{m}(\mathbb{H}[V_i] - \mathbb{H}[X_{i,m}]) \\ &= m\sum_{i=1}^{m} d[X_i; X_i] + \sum_{i=1}^{m} \mathbb{H}[V_i] - \sum_{i=1}^{m} \mathbb{H}[X_i]. \end{aligned} \tag{29}$$

By Theorem 61 (with f taking each j to the index j' such that $X_{i,j}$ is a copy of $X_{j'}$) we obtain the bound

$$\mathbb{H}[V_i] \leq \mathbb{H}[\sum_{j=1}^m X_j] + \sum_{j=1}^m d[X_{i,j}; X_{i,j}].$$

Finally, summing over i and using $D[X_{[m]}] = k$ gives

$$\begin{aligned}\sum_{i=1}^m \mathbb{H}[V_i] - \sum_{i=1}^m \mathbb{H}[X_i] &\leq \sum_{i,j=1}^m d[X_{i,j}; X_{i,j}] + mk \\ &= m\sum_{i=1}^m d[X_i; X_i] + mk,\end{aligned}$$

where in the second step we used the permutation hypothesis. Combining this with (29) gives the

$$\mathcal{I} \leq 2\eta m\left(\sum_{i=1}^m d[X_i; X_i]\right) + mk.$$

The claim (23) is now immediate from Theorem 70. □

13 Endgame

Now let $m \geq 2$, let G be an m-torsion abelian group, and let $(X_i)_{1\leq i\leq m}$ be a τ-minimizer.

Definition 89 (Additional random variables)**.** *By a slight abuse of notation, we identify $\mathbb{Z}/m\mathbb{Z}$ and $\{1, \dots, m\}$ in the obvious way, and let $Y_{i,j}$ be an independent copy of X_i for $i, j \in \mathbb{Z}/m\mathbb{Z}$. Then also define:*

$$W := \sum_{i,j\in\mathbb{Z}/m\mathbb{Z}} Y_{i,j}$$

and

$$Z_1 := \sum_{i,j\in\mathbb{Z}/m\mathbb{Z}} iY_{i,j}, \quad Z_2 := \sum_{i,j\in\mathbb{Z}/m\mathbb{Z}} jY_{i,j}, \quad Z_3 := \sum_{i,j\in\mathbb{Z}/m\mathbb{Z}} (-i-j)Y_{i,j}.$$

The addition $(-i-j)$ takes place over $\mathbb{Z}/m\mathbb{Z}$. Note that, because we are assuming G is m-torsion, it is well-defined to multiply elements of G by elements of $\mathbb{Z}/m\mathbb{Z}$. We will also define for $i, j, r \in \mathbb{Z}/m\mathbb{Z}$ the variables

$$P_i := \sum_{j\in\mathbb{Z}/m\mathbb{Z}} Y_{i,j}, \quad Q_j := \sum_{i\in\mathbb{Z}/m\mathbb{Z}} Y_{i,j}, \quad R_r := \sum_{\substack{i,j\in\mathbb{Z}/m\mathbb{Z}\\ i+j=-r}} Y_{i,j}. \tag{30}$$

Lemma 90 (Zero-sum)**.** *We have*

$$Z_1 + Z_2 + Z_3 = 0 \tag{31}$$

Proof. Clear from definition. □

Proposition 91 (Mutual information bound)**.** *We have*

$$\mathbb{I}[Z_1 : Z_2 \,|\, W],\ \mathbb{I}[Z_2 : Z_3 \,|\, W],\ \mathbb{I}[Z_1 : Z_3 \,|\, W] \leq t$$

where

$$t := m(4m+1)\eta k. \tag{32}$$

Proof. We analyze these variables by Theorem 88 in several different ways. In the first application, take $X_{i,j} = Y_{i,j}$. Note that each column $(X_{i,j})_{i=1}^m$ is indeed a permutation of $X_1, \dots, X_m$; in fact, the trivial permutation. Note also that for each $i \in \mathbb{Z}/m\mathbb{Z}$, the row sum is

$$\sum_{j=1}^m X_{i,j} = \sum_{j \in \mathbb{Z}/m\mathbb{Z}} Y_{i,j} = P_i$$

and for each $j \in \mathbb{Z}/m\mathbb{Z}$, the column sum is

$$\sum_{i=1}^m X_{i,j} = \sum_{i \in \mathbb{Z}/m\mathbb{Z}} Y_{i,j} = Q_j.$$

Finally note that $\sum_{i,j=1}^m X_{i,j} = W$. From Theorem 88 we then have

$$\mathbb{I}[(P_i)_{i \in \mathbb{Z}/m\mathbb{Z}} : (Q_j)_{j \in \mathbb{Z}/m\mathbb{Z}} \,\big|\, W] \leq t,$$

with t as in (32). Since Z_1 is a function of $(P_i)_{i \in \mathbb{Z}/m\mathbb{Z}}$ by (30), and similarly Z_2 is a function of $(Q_j)_{j \in \mathbb{Z}/m\mathbb{Z}}$, it follows immediately from Theorem 56 that

$$\mathbb{I}[Z_1 : Z_2 \,|\, W] \leq t.$$

In the second application of Theorem 88, we instead consider $X'_{i,j} = Y_{i-j,j}$. Again, for each fixed j, the tuple $(X'_{i,j})_{i=1}^m$ is a permutation of $X_1, \dots, X_m$. This time the row sums for $i \in \{1, \dots, m\}$ are

$$\sum_{j=1}^m X'_{i,j} = \sum_{j \in \mathbb{Z}/m\mathbb{Z}} Y_{i-j,j} = R_{-i}.$$

Similarly, the column sums for $j \in \{1, \dots, m\}$ are

$$\sum_{i=1}^m X'_{i,j} = \sum_{i \in \mathbb{Z}/m\mathbb{Z}} Y_{i-j,j} = Q_j.$$

As before, $\sum_{i,j=1}^m X'_{i,j} = W$. Hence, using (30) and Theorem 56 again, Theorem 88 tells us

$$\mathbb{I}[Z_3 : Z_2 \,|\, W] \leq \mathbb{I}[(R_i)_{i \in \mathbb{Z}/m\mathbb{Z}} : (Q_j)_{j \in \mathbb{Z}/m\mathbb{Z}} \,\big|\, W] \leq t.$$

In the third application[1] of Theorem 88, take $X''_{i,j} = Y_{i,j-i}$. The column and row sums are respectively

$$\sum_{j=1}^m X''_{i,j} = \sum_{j \in \mathbb{Z}/m\mathbb{Z}} Y_{i,j-i} = P_i$$

[1] In fact, by permuting the variables $(Y_{i,j})_{i,j \in \mathbb{Z}/m\mathbb{Z}}$, one can see that the random variables (W, Z_1, Z_2) and (W, Z_1, Z_3) have the same distribution, so this is in some sense identical to – and can be deduced from – the first application.

and

$$\sum_{i=1}^{m} X''_{i,j} = \sum_{i \in \mathbb{Z}/m\mathbb{Z}} Y_{i,j-i} = R_{-j}.$$

Hence, Theorem 88 and Theorem 56 give

$$\mathbb{I}[Z_1 : Z_3 \,|\, W] \leq \mathbb{I}[(P_i)_{i \in \mathbb{Z}/m\mathbb{Z}} : (R_j)_{j \in \mathbb{Z}/m\mathbb{Z}} \,\big|\, W] \leq t,$$

which completes the proof. □

Lemma 92 (Entropy of W)**.** *We have* $\mathbb{H}[W] \leq (2m-1)k + \frac{1}{m}\sum_{i=1}^{m} \mathbb{H}[X_i]$.

Proof. Without loss of generality, we may take $X_1, \ldots, X_m$ to be independent. Write $S = \sum_{i=1}^{m} X_i$. Note that for each $j \in \mathbb{Z}/m\mathbb{Z}$, the sum Q_j from (30) above has the same distribution as S. By Theorem 58 we have

$$\begin{aligned}\mathbb{H}[W] = \mathbb{H}[\sum_{j \in \mathbb{Z}/m\mathbb{Z}} Q_j] &\leq \mathbb{H}[S] + \sum_{j=2}^{m} (\mathbb{H}[Q_1 + Q_j] - \mathbb{H}[S]) \\ &= \mathbb{H}[S] + (m-1)d[S; -S].\end{aligned}$$

By Theorem 72, we have

$$d[S; -S] \leq 2k \tag{33}$$

and hence

$$\mathbb{H}[W] \leq 2k(m-1) + \mathbb{H}[S].$$

From Theorem 64 we have

$$\mathbb{H}[S] = k + \frac{1}{m}\sum_{i=1}^{m} \mathbb{H}[X_i], \tag{34}$$

and the claim follows. □

Lemma 93 (Entropy of Z_2)**.** *We have* $\mathbb{H}[Z_2] \leq (8m^2 - 16m + 1)k + \frac{1}{m}\sum_{i=1}^{m} \mathbb{H}[X_i]$.

Proof. We observe

$$\mathbb{H}[Z_2] = \mathbb{H}[\sum_{j \in \mathbb{Z}/m\mathbb{Z}} jQ_j].$$

Applying Theorem 58 one has

$$\mathbb{H}[Z_2] \leq \sum_{i=2}^{m-1} \mathbb{H}[Q_1 + iQ_i] - (m-2)\mathbb{H}[S].$$

Using Theorem 63 and (33) we get

$$\begin{aligned}\mathbb{H}[Z_2] &\leq \mathbb{H}[S] + 4m(m-2)d[S; -S] \\ &\leq \mathbb{H}[S] + 8m(m-2)k.\end{aligned}$$

Applying (34) gives the claim. □

Lemma 94 (Mutual information bound)**.** *We have* $\mathbb{I}[W : Z_2] \leq 2(m-1)k$.

Proof. From Theorem 15 we have $\mathbb{I}[W : Z_2] = \mathbb{H}[W] - \mathbb{H}[W|Z_2]$, and since $Z_2 = \sum_{j=1}^{m-1} jQ_j$ and $W = \sum_{j=1}^{m} Q_j$,

$$\mathbb{H}[W|Z_2] \geq \mathbb{H}[W \,|\, Q_1, \ldots, Q_{m-1}] = \mathbb{H}[Q_m] = \mathbb{H}[S].$$

Hence, by Theorem 92,

$$\mathbb{I}[W : Z_2] \leq \mathbb{H}[W] - \mathbb{H}[S] \leq 2(m-1)k,$$

as claimed. □

Lemma 95 (Distance bound)**.** *We have* $\sum_{i=1}^{m} d[X_i; Z_2|W] \leq 4(m^3 - m^2)k$.

Proof. For each $i \in \{1, \ldots, m\}$, using Theorem 60 (noting the sum Z_2 contains X_i as a summand) we have

$$d[X_i; Z_2] \leq d[X_i; X_i] + \tfrac{1}{2}(\mathbb{H}[Z_2] - \mathbb{H}[X_i]) \tag{35}$$

and using Theorem 47 we have

$$d[X_i; Z_2|W] \leq d[X_i; Z_2] + \tfrac{1}{2}\mathbb{I}[W : Z_2].$$

Combining with (35) and Theorem 94 gives

$$d[X_i; Z_2|W] \leq d[X_i; X_i] + \tfrac{1}{2}(\mathbb{H}[Z_2] - \mathbb{H}[X_i]) + (m-1)k.$$

Summing over i and applying Theorem 93 gives

$$\sum_{i=1}^{m} d[X_i; Z_2|W] \leq \sum_{i=1}^{m} d[X_i; X_i] + m(8m^2 - 16m + 1)k/2 + m(m-1)k.$$

Finally, applying Theorem 70 (and dropping some lower order terms) gives the claim. □

Lemma 96 (Application of BSG)**.** *Let* G *be an abelian group, let* (T_1, T_2, T_3) *be a* G^3*-valued random variable such that* $T_1 + T_2 + T_3 = 0$ *holds identically, and write*

$$\delta := \mathbb{I}[T_1 : T_2] + \mathbb{I}[T_1 : T_3] + \mathbb{I}[T_2 : T_3].$$

Let $Y_1, \ldots, Y_n$ *be some further* G*-valued random variables and let* $\alpha > 0$ *be a constant. Then there exists a random variable* U *such that*

$$d[U; U] + \alpha \sum_{i=1}^{n} d[Y_i; U] \leq \Big(2 + \frac{\alpha n}{2}\Big)\delta + \alpha \sum_{i=1}^{n} d[Y_i; T_2]. \tag{36}$$

Proof. We apply Theorem 46 with $X = T_1$ and $Y = T_2$. Since $T_1 + T_2 = -T_3$, we find that

$$\begin{aligned} &\sum_{z} p_{T_3}(z) d[T_1 \,|\, T_3 = z; T_2 \,|\, T_3 = z] \\ &\qquad \leq 3\mathbb{I}[T_1 : T_2] + 2\mathbb{H}[T_3] - \mathbb{H}[T_1] - \mathbb{H}[T_2] \\ &\qquad = \mathbb{I}[T_1 : T_2] + \mathbb{I}[T_1 : T_3] + \mathbb{I}[T_2 : T_3] = \delta, \end{aligned} \tag{37}$$

where the last line follows from Theorem 5 by observing

$$\mathbb{H}[T_1, T_2] = \mathbb{H}[T_1, T_3] = \mathbb{H}[T_2, T_3] = \mathbb{H}[T_1, T_2, T_3]$$

since any two of T_1, T_2, T_3 determine the third.

By (37) and the triangle inequality,

$$\sum_z p_{T_3}(z) d[T_2 \,|\, T_3 = z; T_2 \,|\, T_3 = z] \le 2\delta$$

and by Theorem 48, for each Y_i,

$$\begin{aligned}&\sum_z p_{T_3}(z) d[Y_i; T_2 \,|\, T_3 = z] \\ &\quad = d[Y_i; T_2 \,|\, T_3] \le d[Y_i; T_2] + \frac{1}{2}\mathbb{I}[T_2 : T_3] \le d[Y_i; T_2] + \frac{\delta}{2}.\end{aligned}$$

Hence,

$$\begin{aligned}&\sum_z p_{T_3}(z) \Big(d[T_2 \,|\, T_3 = z; T_2 \,|\, T_3 = z] + \alpha \sum_{i=1}^n d[Y_i; T_2 \,|\, T_3 = z] \Big) \\ &\quad \le \Big(2 + \frac{\alpha n}{2}\Big)\delta + \alpha \sum_{i=1}^n d[Y_i; T_2],\end{aligned}$$

and the result follows by setting $U = (T_2 \,|\, T_3 = z)$ for some z such that the quantity in parentheses on the left-hand side is at most the weighted average value. □

Proposition 97 (Vanishing entropy)**.** *We have $k = 0$.*

Proof. For each value $W = w$, apply Theorem 96 (and Theorem 90) to

$$T_1 = (Z_1 \,|\, W = w),\ T_2 = (Z_2 \,|\, W = w),\ T_3 = (Z_3 \,|\, W = w)$$

with $Y_i = X_i$ and $\alpha = \eta/m$. Write

$$\delta_w := \mathbb{I}[T_1 : T_2] + \mathbb{I}[T_1 : T_3] + \mathbb{I}[T_2 : T_3]$$

for this choice, and note that

$$\begin{aligned}\delta_* := \sum_w p_W(w)\delta_w &= \mathbb{I}[Z_1 : Z_2 \,|\, W] + \mathbb{I}[Z_1 : Z_3 \,|\, W] + \mathbb{I}[Z_2 : Z_3 \,|\, W] \\ &\le 3m(4m+1)\eta k \qquad (38)\end{aligned}$$

by Theorem 91. Write U_w for the random variable guaranteed to exist by Theorem 96, so that (36) gives

$$d[U_w; U_w] \le \Big(2 + \frac{\alpha m}{2}\Big)\delta_w + \alpha \sum_{i=1}^m \big(d[X_i; T_2] - d[X_i; U_w]\big). \qquad (39)$$

Let $(U_w)_I$ denote the tuple consisting of the same variable U_w repeated m times. By Theorem 71

$$D[(U_w)_I] \leq md[U_w; U_w]. \tag{40}$$

On the other hand, from Theorem 79 one has

$$D[(U_w)_I] \geq k - \eta \sum_{i=1}^{m} d[X_i; U_w]. \tag{41}$$

Combining (39), (40) and (41) and averaging over w (with weight $p_W(w)$), and recalling the value $\alpha = \eta/m$, gives

$$m\Big(2 + \frac{\eta}{2}\Big)\delta_* + \eta \sum_{i=1}^{m} d[X_i; Z_2|W] \geq k$$

since the terms $d[X_i; U_w]$ cancel by our choice of α. Substituting in Theorem 95 and (38), and using the fact that $2 + \frac{\eta}{2} < 3$, we have

$$3m^2(4m+1)(2+\frac{\eta}{2})\eta k + \eta 8(m^3 - m^2)k \geq k.$$

From Theorem 74 we have we have

$$3m^2(4m+1)(2+\frac{\eta}{2})\eta + \eta 8(m^3 - m^2) < 1$$

and hence $k \leq 0$. The claim now follows from Theorem 67. □

14 Wrapping up

Theorem 98 (Entropy form of PFR)**.** *Suppose that G is a finite abelian group of torsion m. Suppose that X is a G-valued random variable. Then there exists a subgroup $H \leq G$ such that*

$$d[X; U_H] \leq 64m^3 d[X; X].$$

Proof. Set $X^0 := X$. By Theorem 77, there exists a τ-minimizer $X_{[m]} = (X_i)_{1 \leq i \leq m}$. By Theorem 97, we have $D[X_{[m]}] = 0$. By Theorem 78 and the pigeonhole principle, there exists $1 \leq i \leq m$ such that $d[X_i; X] \leq \frac{2}{\eta} d[X; X]$. By Theorem 73, we have $d[X_i; U_H] = 0$ for some subgroup $H \leq G$, hence by Theorem 41 we have $d[U_H; X] \leq \frac{2}{\eta} d[X; X]$. The claim then follows from Theorem 74. □

Lemma 99. *Suppose that G is a finite abelian group of torsion m. If $A \subset G$ is non-empty and $|A + A| \leq K|A|$, then A can be covered by at most $K^{128m^3+1}|A|^{1/2}/|H|^{1/2}$ translates of a subspace H of G with*

$$|H|/|A| \in [K^{-256m^3}, K^{256m^3}]. \tag{42}$$

Proof. Repeat the proof of Theorem 2, but with Theorem 98 in place of Theorem 52. Because of the lack of 2-torsion, one has to use the Ruzsa triangle inequality to bound $d[U, U]$ by $2d[U, -U]$, costing an additional factor of 2 in the estimates. □

Lemma 100 (Symmetry group is a group)**.** *If X is a G-valued random variable, then* $\mathrm{Sym}[X]$ *is a subgroup of G.*

Proof. Direct verification of the group axioms. □

Lemma 101 (Translate is uniform on symmetry group)**.** *If X is a G-valued random variable with $d[X;X] = 0$, and x_0 is a point with $P[X = x_0] > 0$, then $X - x_0$ is uniformly distributed on* $\mathrm{Sym}[X]$.

Proof. The law of $X - x_0$ is invariant under $\mathrm{Sym}[X]$, non-zero at the origin, and supported on $\mathrm{Sym}[X]$, giving the claim. □

Lemma 102 (Symmetric 100% inverse theorem)**.** *Suppose that X is a G-valued random variable such that $d[X;X] = 0$. Then there exists a subgroup $H \leq G$ such that $d[X;U_H] = 0$.*

Proof. Take H to be the symmetry group of X, which is a group by Theorem 100. From Theorem 101, $X - x_0$ is uniform on H, and $d[X;X - x_0] = d[X;X] \leq 0$, and the claim follows. □

Corollary 103 (General 100% inverse theorem)**.** *Suppose that X_1, X_2 are G-valued random variables such that $d[X_1;X_2] = 0$. Then there exists a subgroup $H \leq G$ such that $d[X_1;U_H] = d[X_2;U_H] = 0$.*

Proof. Using Theorem 41 and Theorem 39 we have $d[X_1;X_1] = 0$, hence by Theorem 102 $d[X_1;U_H] = 0$ for some subgroup H. By Theorem 41 and Theorem 39 again we also have $d[X_2;U_H]$ as required. □

Theorem 104 (PFR)**.** *Suppose that G is an abelian group of torsion m (not necessarily finite). If $A \subset G$ is a finite non-empty set with $|A + A| \leq K|A|$, then A can be covered by at most mK^{256m^3+1} translates of a finite subspace H of G with $|H| \leq |A|$.*

Proof. If G is finite, repeat the proof of Theorem 3, but with Theorem 99 in place of Theorem 2. In general, reduce to the finite span of A over $\mathbb{Z}/m\mathbb{Z}$. □